Chapter 27: Wave Properties of Particles

de Broglie hypothesis:

The de Broglie hypothesis, proposed by the French physicist Louis de Broglie in 1924, suggests that particles, such as electrons, can exhibit both wave-like and particle-like behavior.

De Broglie's hypothesis is based on the idea that if light, Which was previously thought to be purely wave-like in nature, can Exhibit particle-like behavior (as demonstrated by the photoelectric effect), then particles like electrons, which were traditionally considered to be particles, might also exhibit wave-like properties.

De Broglie postulated that every moving particle, with momentum p, is associated with a wave whose wavelength (λ) is related to the particle's momentum by the equation:

$$\lambda = \frac{h}{P}$$

λ is the wavelength of the associated wave.
h is the Planck constant (a fundamental constant of nature).
p is the linear momentum of the particle.

NOTE: The linear momentum is P = mV, m is the particle Mass and V is its velocity.
The kinetic energy of the particle is $k = \frac{1}{2}mV^2 = \frac{P^2}{2m}$

Example is the electron diffraction experiment

- Check questions 27.7, 27.8 and 27.12 of set #(8)

Wave-Particle Duality:

Wave-Particle Duality helps us to understand the particle and wave nature of light. Based on the idea that light and all other electromagnetic radiation may be considered a particle or a wave nature, in 1923 physicists Louis De Broglie suggested that the same kind of duality must apply to the matter

Summary:

In these two chapters (26 & 27) we focused on the particle behavior of light. We give two examples: the photoelectric effect and Compton scattering. Furthermore, our attention is directed towards the wave-like characteristics of particles, as postulated by de Broglie. An example of this is the electron diffraction experiment.

- Photoelectric effect $\rightarrow$ Particle nature of light
- Compton effect $\rightarrow$ Particle nature of light
- Electron diffraction $\rightarrow$ Wave nature of particles
- Diffraction and interference of light $\rightarrow$ Wave nature of light